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Article

A Collatz-Equivalent Map on the Nonzero Integers

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Abstract

The graph of the accelerated Collatz map T carries structure that is irrelevant to the Collatz conjecture: one third of its vertices are multiples of three, which no orbit can revisit, and the negative integers cannot be used at all because T admits non-trivial cycles on them. We construct an explicit conjugacy J between the conjecture-relevant vertices $[1]_3 \cup [2]_3$ and the nonzero integers $\mathbb{Z}^*$, yielding a map $K: \mathbb{Z}^* \rightarrow \mathbb{Z}^*$ whose graph is isomorphic to the pruned Collatz graph. The Collatz conjecture becomes the statement that every K -orbit reaches the two-cycle $\{1, -1\}$; in these coordinates the sign of an iterate records its residue class modulo 3, so every nonzero integer indexes a conjecture-relevant vertex and no non-trivial cycles are introduced. We analyze the edge structure of the new graph, identify a spanning acyclic subgraph that is bipartite under the sign 2-coloring, and introduce an accelerated map $\hat{K}$ whose graph is bipartite outright and whose in-degree sequence is governed by the 3-adic valuation $v_3(2k - 1)$, recovering OEIS A254046 with mean in-degree $3/2$. Finally, we observe that K -orbits, plotted as signed time series, behave like damped seismic signals: they oscillate across zero with limiting sign-change frequency $2/3$ while their envelope decays at the heuristic rate $\lambda = \frac{1}{2} \ln \frac{4}{3}$ per step, suggesting dynamical and spectral tools as instruments for studying Collatz dynamics. No proof of the conjecture is claimed; the aim is a coordinate system in which its dynamics are easier to see.

Keywords: Collatz conjecture; graph theory; seismic graphs; number theory

1. Introduction

Let $\mathbb{Z}^+$ denote the positive integers. The Collatz function is

$$C(n) = \begin{cases} 3n + 1 & n \text{ odd,} \\ n/2 & n \text{ even,} \end{cases} \quad n \in \mathbb{Z}^+. \tag{1}$$

Conjecture 1 (Collatz). *For every $n \in \mathbb{Z}^+$ the orbit $n, C(n), C(C(n)), \dots$ eventually enters the cycle $4 \rightarrow 2 \rightarrow 1 \rightarrow 4$.*

Since $3n + 1$ is even whenever n is odd, Terras [1] combined the odd step with the division that necessarily follows it, producing the *accelerated* (or *reduced*) map

$$T(n) = \begin{cases} \frac{3n + 1}{2} & n \text{ odd,} \\ \frac{n}{2} & n \text{ even,} \end{cases} \quad n \in \mathbb{Z}^+, \tag{2}$$

for which the trivial cycle becomes $2 \rightarrow 1 \rightarrow 2$. The conjecture for T is equivalent to Conjecture 1, and we work with T throughout. We write G_T for the functional graph of T on $\mathbb{Z}^+$: the directed graph with vertex set $\mathbb{Z}^+$ and an edge $n \rightarrow T(n)$ for every n . For the extensive literature on the problem see Lagarias [2,3].

Two features of G_T limit how faithfully it represents the conjecture it encodes.

First, *a third of its vertices are inert*. No orbit can enter the multiples of three: as recalled in Section 3, an orbit that starts in $[0]_3$ leaves that residue class after finitely many steps and never returns, while an orbit that starts outside $[0]_3$ never visits it. Consequently the multiples of three are irrelevant to the existence of cycles or divergent orbits, yet they occupy one third of the vertex set of G_T .

Second, *the negative integers are unusable*. Extending C or T to $\mathbb{Z}^-$ produces well-known non-trivial cycles, for instance

$$-1 \rightarrow -2 \rightarrow -1, \quad -5 \rightarrow -14 \rightarrow -7 \rightarrow -20 \rightarrow -10 \rightarrow -5,$$

and the length-18 cycle through -17 (equivalently, these are the known cycles of the $3x - 1$ problem on $\mathbb{Z}^+$; see [2]). The negative half of $\mathbb{Z}$ therefore cannot serve as extra room for encoding Collatz dynamics—at least not for the maps C and T themselves.

This paper addresses both limitations at once. We exhibit an explicit bijection J from the conjecture-relevant residue classes $[1]_3 \cup [2]_3$ onto the nonzero integers $\mathbb{Z}^* = \mathbb{Z}^+ \cup \mathbb{Z}^-$, and we conjugate T by J to obtain a map $K: \mathbb{Z}^* \rightarrow \mathbb{Z}^*$,

$$K(n) = \begin{cases} \frac{1-3n}{2} & n > 0 \text{ odd}, \\ -\frac{n}{2} & n > 0 \text{ even}, \\ \frac{1-n}{2} & n < 0 \text{ odd}, \\ \frac{3n}{2} & n < 0 \text{ even}, \end{cases} \quad (3)$$

whose functional graph G_K is isomorphic to the graph obtained from G_T by pruning the multiples of three. Under this conjugacy:

- every nonzero integer indexes a conjecture-relevant vertex — the graph has no inert third;
- the sign of an iterate records its residue class: positive integers correspond to $[1]_3$ and negative integers to $[2]_3$;
- no new cycles appear — the Collatz conjecture is equivalent to the statement that every K -orbit reaches the two-cycle $\{1, -1\}$ (Theorem 6).

The map K thus realizes, by construction, the property one might wish T had: a single map on *all* nonzero integers with (conjecturally) a single cycle.

The paper is organized as follows. Section 2 fixes notation and records how T permutes residue classes modulo 3. Section 3 formalizes the pruning of $[0]_3$ and proves that the pruned conjecture is equivalent to the original. Section 4 constructs J and K and proves the conjugacy. Section 5 decomposes the edges of G_K , exhibits a spanning acyclic subgraph that is bipartite under the sign 2-coloring, and explains both why a rule-based 2-coloring of the full graph appears out of reach and why bipartiteness alone could not settle the conjecture. Section 6 introduces an accelerated map $\hat{K}$ that contracts runs through negative even integers; its graph is bipartite outright, and its in-degree arithmetic connects to the OEIS sequences A254046, A191450, A087289 and A036561. Section 7 develops the dynamical reading that motivated the construction: K -orbits, plotted as signed time series, are damped irregular oscillations, quantitatively comparable to the decaying coda of a seismogram, and we propose this as a framework for importing tools from the theory of random dynamical systems and signal processing. Section 8 concludes. Closed forms and reference implementations of K appear in Appendix A.

We emphasize the scope of the contribution: no progress on the truth of Conjecture 1 is claimed. The contribution is a pair of equivalent reformulations—one on $\mathbb{Z}^*$, one back on $\mathbb{Z}^+$ via acceleration—together with structural results about their graphs, intended as a more transparent coordinate system for future investigations.

2. Preliminaries: The Action of T on Residue Classes Mod 3

Notation 2. $\mathbb{Z}^+ = \{1, 2, 3, \dots\}$, $\mathbb{Z}^- = \{-1, -2, -3, \dots\}$, and $\mathbb{Z}^* = \mathbb{Z}^+ \cup \mathbb{Z}^-$. For r, m integers with $m \geq 2$ we write $[r]_m = \{n \in \mathbb{Z}^+ : n \equiv r \pmod{m}\}$; thus $[0]_3, [1]_3, [2]_3$ partition $\mathbb{Z}^+$. We write $v_p(n)$ for

the p -adic valuation of $n \neq 0$ (the exponent of the prime p in n), and $f^{(t)}$ for the t -fold iterate of a map f . The orbit of n under f is the sequence $(f^{(t)}(n))_{t \geq 0}$.

Splitting each residue class modulo 3 by parity gives the six classes modulo 6, on which the action of T is completely described by the following computation.

Lemma 1 (Residue transitions). *Writing each class of $\mathbb{Z}^+$ modulo 6 as $\{6k + r\}_{k \geq 0}$ (omitting $k = 0$ when $r = 0$), the map T acts as follows:*

class	parity	image of $6k + r$	image class
$[0]_6$	even	$3k$	$[0]_3$
$[3]_6$	odd	$9k + 5$	$[5]_9 \subset [2]_3$
$[4]_6$	even	$3k + 2$	$[2]_3$ (onto)
$[1]_6$	odd	$9k + 2$	$[2]_9 \subset [2]_3$
$[2]_6$	even	$3k + 1$	$[1]_3$ (onto)
$[5]_6$	odd	$9k + 8$	$[8]_9 \subset [2]_3$

In particular $T(n) \in [0]_3$ if and only if $n \in [0]_6$, and $T([1]_3 \cup [2]_3) \subseteq [1]_3 \cup [2]_3$.

Proof. Direct computation from (2). For example, if $n = 6k + 1$ then n is odd and $T(n) = (18k + 4)/2 = 9k + 2$; the other five cases are identical in kind. $\square$

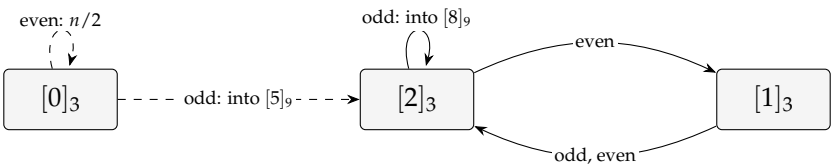


Figure 1. The action of T on residue classes modulo 3 (Lemma 1). Dashed arrows involve $[0]_3$, which no orbit can enter from outside; solid arrows form the conjecture-relevant part of the dynamics.

3. Pruning the Multiples of Three

That multiples of three form a transient, inert part of the Collatz graph is folklore; we record the statements we need in a form suited to the conjugacy of Section 4.

Lemma 2. *Let $n \in \mathbb{Z}^+$ and consider the T -orbit of n .*

- 1. *If $n \notin [0]_3$, the orbit never visits $[0]_3$.*
- 2. *If $n \in [0]_3$, the orbit consists of an initial segment inside $[0]_3$ of finite length at most $v_2(n) + 1$, after which it enters $[1]_3 \cup [2]_3$ and remains there forever.*

Proof. (1) By Lemma 1, the only class mapping into $[0]_3$ is $[0]_6 \subset [0]_3$. (2) While the orbit stays in $[0]_3$, each element is either even—in which case T halves it, strictly decreasing its 2-adic valuation—or odd, in which case Lemma 1 sends it into $[5]_9 \subset [2]_3$. An odd multiple of 3 is reached after at most $v_2(n)$ halvings, so the orbit leaves $[0]_3$ after at most $v_2(n) + 1$ steps, and by (1) it never returns. $\square$

Definition 1. *For a directed graph G with vertex set $V(G) \subseteq \mathbb{Z}^+$, let $P(G)$ be the graph obtained by deleting the vertices in $[0]_3$ together with all incident edges. Write $G_{TP} := P(G_T)$. Equivalently, G_{TP} is the functional graph of the restriction $T: [1]_3 \cup [2]_3 \rightarrow [1]_3 \cup [2]_3$, which is well defined by Lemma 1.*

Conjecture 3. *For every $n \in [1]_3 \cup [2]_3$, the T -orbit of n reaches the cycle $\{1, 2\}$.*

Theorem 4. *Conjecture 3 is equivalent to the Collatz conjecture.*

Proof. Conjecture 1 (for T) trivially implies Conjecture 3. Conversely, assume Conjecture 3 and let $n \in \mathbb{Z}^+$. If $n \notin [0]_3$ we are done. If $n \in [0]_3$, then by Lemma 2 its orbit reaches some $m \in [1]_3 \cup [2]_3$ in finitely many steps, and the orbit of m reaches $\{1, 2\}$ by assumption. Hence every T -orbit reaches $\{1, 2\}$; cycles and divergent orbits of T , if any, must therefore live entirely inside G_{TP} . $\square$

Thus G_{TP} is exactly the part of the Collatz graph in which cycles or divergent orbits can reside: it retains all of the difficulty of the conjecture while shedding one third of the vertex set.

4. The Conjugated Map K on the Nonzero Integers

The two classes $[1]_3$ and $[2]_3$ are each countable, and we now put them in explicit bijection with $\mathbb{Z}^+$ and $\mathbb{Z}^-$ respectively.

Definition 2. Define $J: [1]_3 \cup [2]_3 \rightarrow \mathbb{Z}^*$ and its inverse by

$$J(n) = \begin{cases} \frac{n+2}{3} & n \in [1]_3, \\ -\frac{n+1}{3} & n \in [2]_3, \end{cases} \quad J^{-1}(m) = \begin{cases} 3m-2 & m > 0, \\ -3m-1 & m < 0. \end{cases} \quad (4)$$

Thus J sends $1, 4, 7, 10, \dots$ to $1, 2, 3, 4, \dots$ and $2, 5, 8, 11, \dots$ to $-1, -2, -3, -4, \dots$; it is a bijection with the stated inverse by direct computation.

Theorem 5 (Conjugacy). Let $K: \mathbb{Z}^* \rightarrow \mathbb{Z}^*$ be defined by (3). Then

$$K = J \circ T \circ J^{-1}.$$

Consequently the functional graph G_K of K on $\mathbb{Z}^*$ is isomorphic to G_{TP} , via the vertex bijection J .

Proof. We verify the four cases of (3).

Case $m > 0$ odd. Write $m = 2j + 1$. Then $n = J^{-1}(m) = 3m - 2 = 6j + 1$ is odd, so $T(n) = 9j + 2 \in [2]_9 \subset [2]_3$ by Lemma 1, and $J(T(n)) = -(9j + 3)/3 = -(3j + 1) = \frac{1-3m}{2}$.

Case $m > 0$ even. Write $m = 2j$. Then $n = 6j - 2$ is even, $T(n) = 3j - 1 \in [2]_3$, and $J(T(n)) = -j = -m/2$.

Case $m < 0$ odd. Write $m = -(2j + 1)$, $j \geq 0$. Then $n = -3m - 1 = 6j + 2$ is even, $T(n) = 3j + 1 \in [1]_3$, and $J(T(n)) = j + 1 = \frac{1-m}{2}$.

Case $m < 0$ even. Write $m = -2j$, $j \geq 1$. Then $n = 6j - 1$ is odd, $T(n) = 9j - 1 \in [8]_9 \subset [2]_3$, and $J(T(n)) = -3j = \frac{3m}{2}$.

In each case $J(T(J^{-1}(m)))$ agrees with (3). Since J is a bijection intertwining the two maps, it carries edges of G_{TP} to edges of G_K bijectively. $\square$

Remark 1. Restricted to positive inputs, (3) reads $K(m) = -U(m)$, where $U(m) = (3m - 1)/2$ for m odd and $U(m) = m/2$ for m even is the accelerated $3x - 1$ map. This is consistent with the classical observation that the $3x - 1$ problem on $\mathbb{Z}^+$ is the $3x + 1$ problem on $\mathbb{Z}^-$ [2]: in K -coordinates the sign flip that accompanies each application of U is precisely what prevents the known $3x - 1$ cycles from closing up, threading positive and negative integers into a single system with (conjecturally) one cycle.

Theorem 6 (Equivalent conjecture). The following statement is equivalent to the Collatz conjecture: for every $n \in \mathbb{Z}^*$, the K -orbit of n reaches the two-cycle $1 \rightarrow -1 \rightarrow 1$.

Proof. By Theorem 5, J conjugates T on $[1]_3 \cup [2]_3$ with K on $\mathbb{Z}^*$, and $J(\{1, 2\}) = \{1, -1\}$. Hence every K -orbit reaches $\{1, -1\}$ if and only if every T -orbit in G_{TP} reaches $\{1, 2\}$, which is Conjecture 3, which is equivalent to the Collatz conjecture by Theorem 4. $\square$

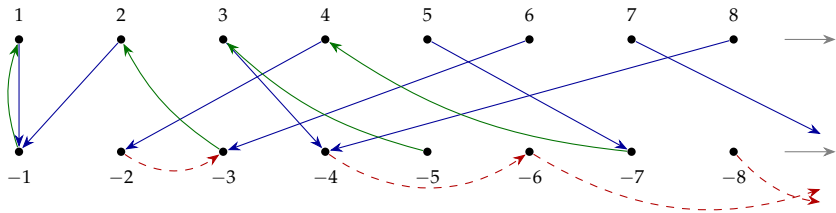


Figure 2. The graph G_K for $|n| \leq 8$, drawn on two parallel rays. Blue edges (E_1): every positive integer maps to a negative one. Green edges (E_2): odd negative integers map back to positive ones. Dashed red edges (E_3): even negative integers map to negative multiples of 3. If the two rays are pictured as the eyelets of a shoe, the Collatz conjecture states that a lace threaded forward from any eyelet reaches the eyelet labeled 1.

Example 7. Iterating K from 20:

$$20, -10, -15, 8, -4, -6, -9, 5, -7, 4, -2, -3, 2, -1, 1, -1, \dots$$

Applying J^{-1} termwise recovers the T -orbit of $J^{-1}(20) = 58$:

$$58, 29, 44, 22, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1, 2, \dots$$

Example 8. The number 27, famous for its long orbit, is a multiple of 3 and hence pruned; the first conjecture-relevant element of its orbit is $T(27) = 41$, with $J(41) = -14$. The K -orbit of -14 (length 70 to reach 1) is the image of the T -orbit of 41 under J :

$$-14, -21, 11, -16, -24, -36, -54, -81, 41, -61, 31, -46, \dots, 4, -2, -3, 2, -1, 1.$$

Both orbits are plotted in Section 7 (Figures 4 and 5).

Remark 2. The apparent complexity of (3)—four branches instead of two—is the price of folding two residue classes into one signed line. What is bought: G_K has the same vertex set as the conjecture-relevant subgraph of G_T , but that vertex set is now all of $\mathbb{Z}^*$, with the residue-class information carried by the sign rather than hidden in the vertex labels. Membership of an iterate in $[1]_3$ or $[2]_3$ —invisible in a plot of a T -orbit—is immediately legible in a plot of a K -orbit.

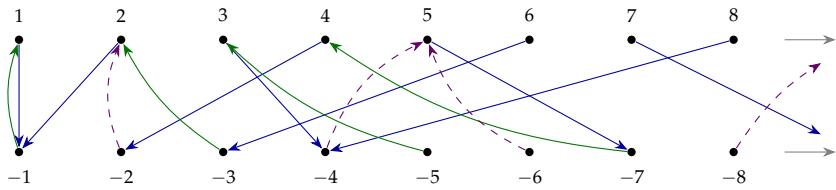


Figure 3. The graph $G_{\hat{K}}$ for $|n| \leq 8$. The red intra-negative edges of Figure 2 are replaced by dashed violet edges from negative even integers directly to positive integers (e.g. $\hat{K}(-4) = \hat{K}(-6) = 5$); every edge now crosses between the rays, so the sign coloring is proper.

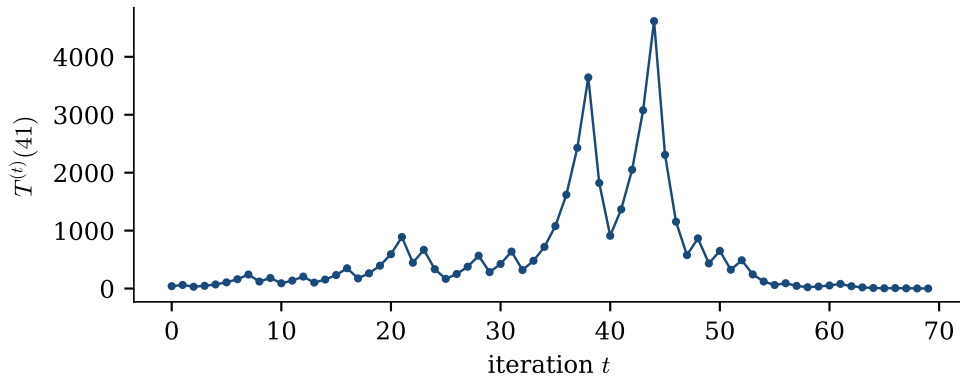


Figure 4. The T -orbit of 41: the classical hailstone picture. Residue-class information is invisible.

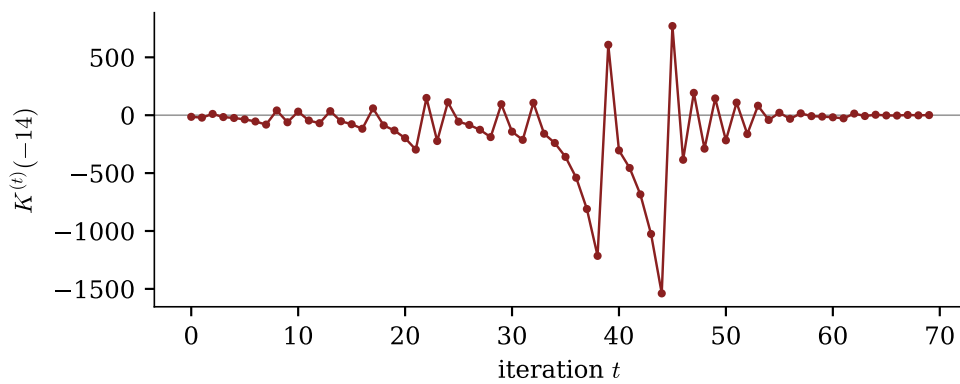


Figure 5. The same orbit in K -coordinates (the K -orbit of $J(41) = -14$). Sign changes mark transitions between $[1]_3$ (positive) and $[2]_3$ (negative); the orbit is a damped irregular oscillation terminating in the ambient two-cycle $\{1, -1\}$.

5. Edge Decomposition and Two-Colorings

Definition 3. Partition the edges of G_K by the branch of (3) that generates them:

$$\begin{aligned} E_1 &= \{ n \rightarrow K(n) : n \in \mathbb{Z}^+ \} && (\text{positive} \rightarrow \text{negative}), \\ E_2 &= \{ n \rightarrow K(n) : n \in \mathbb{Z}^-, n \text{ odd} \} && (\text{negative odd} \rightarrow \text{positive}), \\ E_3 &= \{ n \rightarrow K(n) : n \in \mathbb{Z}^-, n \text{ even} \} && (\text{negative even} \rightarrow \text{negative}). \end{aligned}$$

That the images are as indicated follows from (3): both positive branches have negative values, the negative-odd branch is positive, and the negative-even branch $3n/2$ stays negative. Define $G_K^{1,2} := G_K - E_3$, the spanning subgraph retaining only $E_1 \cup E_2$.

Proposition 1. $G_K^{1,2}$ is bipartite: the 2-coloring $\{\mathbb{Z}^+ \mid \mathbb{Z}^-\}$ (“blue if positive, red if negative”) is proper. In particular $G_K^{1,2}$ contains no odd cycle.

Proof. Every edge of E_1 goes from a positive to a negative vertex and every edge of E_2 from a negative to a positive vertex, so no edge joins two vertices of the same sign. A cycle in a bipartite graph alternates sides and hence has even length. $\square$

The subgraph $G_K^{1,2}$ is not merely bipartite; removing E_3 destroys *every* potential non-trivial cycle and divergent orbit. The proof uses a classical style of minimum-element argument, phrased here in T -coordinates where magnitudes are natural. Note first that under J , the odd elements of $[2]_3$ —that

is, $[5]_6 = \{5, 11, 17, \dots\}$ —correspond exactly to the negative even integers, so E_3 is the image of the edges of G_{TP} emanating from $[5]_6$.

Lemma 3. Suppose the T -orbit $\mathcal{O} = (n_0, n_1, n_2, \dots)$ of some $n_0 \in [1]_3 \cup [2]_3$ is a non-trivial cycle or is divergent, and let $n = \min \mathcal{O}$ (≥ 3 , since orbits meeting $\{1, 2\}$ enter the trivial cycle). Then n is odd, $T(n)$ is odd, and $T(n) \in [5]_6$; moreover if $n \in [5]_6$ we may take the element of $[5]_6$ to be n itself. In particular $\mathcal{O}$ visits $[5]_6$ within two steps of its minimum.

Proof. If n were even, $T(n) = n/2 < n$ would contradict minimality; so n is odd and $m := T(n) = (3n + 1)/2$. If m were even, then $T(m) = (3m + 1)/4 < n$, again contradicting minimality; so m is odd. Since n is odd and not a multiple of 3, either $n \in [5]_6$ (and we are done), or $n \in [1]_6$. In the latter case write $n = 6x + 1$; then $m = 9x + 2$, and m odd forces x odd, say $x = 2z + 1$, giving $m = 18z + 11 \equiv 5 \pmod{6}$. $\square$

Theorem 9. In $G_K^{1,2}$ every negative even vertex is a sink, every orbit either terminates at a negative even vertex or reaches the cycle $\{1, -1\}$, and $\{1, -1\}$ is the unique cycle. In particular $G_K^{1,2}$ has no divergent orbits and no non-trivial cycles.

Proof. Negative even vertices lose their unique outgoing edge (which belonged to E_3), so they are sinks. Suppose some orbit of $G_K^{1,2}$ were a non-trivial cycle or divergent. Then it would avoid all sinks, hence avoid negative even vertices entirely, hence be a non-trivial cycle or divergent orbit of K itself avoiding the negative evens. Its image under J^{-1} would be a non-trivial cycle or divergent T -orbit avoiding $[5]_6$. But by Lemma 3 any such orbit visits $[5]_6$ —a contradiction. $\square$

Remark 3 (Why a rule-based 2-coloring of G_K is elusive, and what it would be worth). Restoring E_3 breaks the sign coloring: E_3 joins negative vertices to negative vertices. Assuming the Collatz conjecture, G_K is a tree together with the 2-cycle $\{1, -1\}$ attached at its root, and trees are bipartite; so conjecturally a proper 2-coloring of G_K exists. The natural way to produce one is to walk the tree from the root, alternating colors level by level—but the levels of the Collatz tree are precisely the orbit lengths whose behavior is the content of the conjecture, so this is circular. A closed-form coloring rule (one computable from the residue data of n alone, without iterating) would be a structural statement about all of G_K at once. We record the modest and the honest version of this observation:

1. Bipartiteness of G_K would rule out non-trivial cycles of odd length only; bipartite graphs admit even cycles, so a 2-coloring cannot by itself prove the conjecture.
2. Conversely, exhibiting an odd cycle in G_K would disprove the Collatz conjecture.

Conjecture 10. G_K is bipartite. (Implied by, and strictly weaker than, the Collatz conjecture.)

6. The Accelerated Map $\hat{K}$ and the Arithmetic of Parents

The obstruction to the sign coloring is the edge set E_3 : runs of the orbit through negative even integers. These runs are completely understood, and contracting them yields a map whose graph is bipartite outright.

Lemma 4. Let $n \in \mathbb{Z}^-$ be even and let $v = v_2(n)$. Then

$$K^{(v)}(n) = n \left(\frac{3}{2} \right)^v,$$

which is a negative odd multiple of 3 with $v_3(K^{(v)}(n)) = v_2(n) + v_3(n)$, and

$$K^{(v+1)}(n) = \frac{1 + |n| (3/2)^v}{2} \in \mathbb{Z}^+.$$

Proof. For negative even m , $K(m) = 3m/2$ satisfies $v_2(K(m)) = v_2(m) - 1$ and $v_3(K(m)) = v_3(m) + 1$. Iterating v times multiplies by $(3/2)^v$ and exhausts the 2-adic valuation, leaving a negative odd number $m' = n(3/2)^v$ with $v_3(m') = v_2(n) + v_3(n)$. One further application of the negative-odd branch gives $K(m') = (1 - m')/2 = (1 + |n|(3/2)^v)/2 > 0$. $\square$

The intermediate quantity $n(3/2)^v$ is not injective in n (for example both -8 and -12 reach -27) nor monotone (compare $-8 \mapsto -27$ with $-10 \mapsto -15$); the runs through negative evens carry genuine arithmetic, which the following map contracts.

Definition 4. Define $\hat{K}: \mathbb{Z}^* \rightarrow \mathbb{Z}^*$ by

$$\hat{K}(n) = \begin{cases} K(n) & n > 0, \text{ or } n < 0 \text{ odd,} \\ \frac{1 + |n|(3/2)^{v_2(n)}}{2} & n < 0 \text{ even,} \end{cases}$$

i.e. $\hat{K}$ agrees with K except that from a negative even integer it jumps directly to the positive integer that K reaches after $v_2(n) + 1$ steps (Lemma 4). Write $G_{\hat{K}}$ for its functional graph.

Proposition 2. $G_{\hat{K}}$ contains a non-trivial cycle if and only if G_K does, and a divergent orbit if and only if G_K does. Consequently the Collatz conjecture is also equivalent to: every $\hat{K}$ -orbit reaches $\{1, -1\}$.

Proof. Every $\hat{K}$ -orbit is obtained from the corresponding K -orbit by deleting the interior of each maximal run through negative even integers; each such run is finite (of length $v_2(n)$, Lemma 4). Deleting finitely many consecutive elements between fixed endpoints neither creates nor destroys eventual periodicity or divergence, and the trivial cycles correspond. $\square$

Theorem 11. Consecutive $\hat{K}$ -iterates have opposite signs. Hence $G_{\hat{K}}$ is bipartite under the sign coloring, and every cycle of $G_{\hat{K}}$ has even length.

Proof. By (3), $\hat{K}$ maps positive integers to negative ones and odd negative integers to positive ones; by Definition 4 it maps even negative integers to positive ones. So every step changes sign. $\square$

Corollary 1. In any hypothetical non-trivial cycle of T (within G_{TP}), the number of elements congruent to 5 modulo 6 has the same parity as the length of the cycle.

Proof. Transport the cycle to G_K by J ; elements of $[5]_6$ correspond to negative even integers. Contracting the negative-even runs yields a cycle of $G_{\hat{K}}$, whose length—the original length minus the number of negative even elements—is even by Theorem 11. $\square$

6.1. Parents and In-degrees

Because $\hat{K}$ is a function, the preimage sets $\hat{K}^{-1}(k)$, $k \in \mathbb{Z}^*$, partition $\mathbb{Z}^*$; the arithmetic of these sets turns out to be governed by the 3-adic valuation of $2k - 1$.

Proposition 3 (Parent formula). Let $k \in \mathbb{Z}^+$. The preimages of k under $\hat{K}$ are:

1. the odd negative integer $-(2k - 1)$, always; and
2. the even negative integers

$$\pi(k) = \left\{ -2^a \frac{2k-1}{3^a} : 1 \leq a \leq v_3(2k-1) \right\},$$

which is nonempty precisely when $k \equiv 2 \pmod{3}$.

Hence the in-degree of k in $G_{\hat{K}}$ is $1 + v_3(2k - 1)$.

Proof. A negative odd m satisfies $\widehat{K}(m) = (1 - m)/2 = k$ iff $m = -(2k - 1)$. A negative even n with $v = v_2(n)$ satisfies $\widehat{K}(n) = k$ iff $|n|(3/2)^v = 2k - 1$, i.e. $|n| = 2^v(2k - 1)/3^v$. This is an integer of 2-adic valuation exactly v iff $3^v \mid 2k - 1$ (the cofactor $(2k - 1)/3^v$ is automatically odd), giving one parent for each $a = 1, \dots, v_3(2k - 1)$. Finally $v_3(2k - 1) \geq 1$ iff $2k \equiv 1 \pmod{3}$ iff $k \equiv 2 \pmod{3}$. $\square$

Example 12. $\pi(5) = \{-6, -4\}$ (since $2 \cdot 5 - 1 = 9 = 3^2$); $\pi(14) = \{-18, -12, -8\}$ (since $27 = 3^3$); $\pi(122) = \{-162, -108, -72, -48, -32\}$ (since $243 = 3^5$). Adding the odd parents $-9, -27, -243$ respectively gives in-degrees 3, 4, 6.

Proposition 4 (Connection to OEIS A254046). For $y \geq 1$ let $k_y = 3y - 1$ enumerate $[2]_3$. Then

$$|\pi(k_y)| = 1 + v_3(2y - 1) = v_3(2^{2y-1} + 1),$$

which is the OEIS sequence A254046 [7], the column index of y in the dispersion of $3n - 1$ (A191450 [8]); the last equality also identifies it as the 3-adic valuation of A087289 [9].

Proof. $2k_y - 1 = 6y - 3 = 3(2y - 1)$, so $|\pi(k_y)| = v_3(2k_y - 1) = 1 + v_3(2y - 1)$ by Proposition 3. For the second equality, the lifting the exponent lemma applies to $2^{2y-1} + 1^{2y-1}$ since $3 \mid 2 + 1$ and $2y - 1$ is odd: $v_3(2^{2y-1} + 1) = v_3(2 + 1) + v_3(2y - 1) = 1 + v_3(2y - 1)$. The first 41 values, 1, 2, 1, 1, 3, 1, 1, 2, 1, 1, 2, 1, 1, 4, $\dots$, 5, match A254046. $\square$

Proposition 5 (Mean number of even parents). The average of $|\pi(k)|$ over $k \in [2]_3$ exists and equals

$$\lim_{Y \rightarrow \infty} \frac{1}{Y} \sum_{y=1}^Y |\pi(3y - 1)| = 1 + \sum_{j \geq 1} 3^{-j} = \frac{3}{2}.$$

Proof. $|\pi(3y - 1)| = 1 + v_3(2y - 1) = 1 + \sum_{j \geq 1} 1[3^j \mid 2y - 1]$, and for each j the set $\{y : 3^j \mid 2y - 1\}$ is an arithmetic progression of density 3^{-j} . Summing the (dominated) series gives $1 + \frac{1/3}{1-1/3} = \frac{3}{2}$. (Numerically, the average over $y \leq 10^5$ is 1.50001.) $\square$

Remark 4. The extremal vertices are transparent: $k = (3^a + 1)/2$ has $2k - 1 = 3^a$ and hence in-degree $a + 1$, the maximum possible for its size. Its even parents $\{-2 \cdot 3^{a-1}, -4 \cdot 3^{a-2}, \dots, -2^a\}$, read together with the odd parent -3^a , form a row of the Nicomachus triangle $\{2^i 3^j\}$ (A036561 [10]) up to sign: the contracted negative-even runs interpolate geometrically between a power of 2 and a power of 3. For example $41 = (3^4 + 1)/2$ has parents $\{-81, -54, -36, -24, -16\}$.

Remark 5 (Growth bound). Lemma 4 bounds the single-step growth of $\widehat{K}$: for negative even n , $|\widehat{K}(n)| = \frac{1}{2}(1 + |n|(3/2)^{v_2(n)})$, so large jumps require large 2-adic valuation—e.g. $n = -160 = -2^5 \cdot 5$ jumps to $\widehat{K}(n) = 608$. All other branches contract or grow by a factor at most $3/2$. In $G_{\widehat{K}}$ the “hailstone” surges of the classical map are thus exactly indexed by the 2-adic valuations along the orbit.

7. K-Orbits as Damped Oscillations: A Seismic Reading

We now develop the dynamical picture that motivated the construction. Everything in this section is either elementary, empirical, or explicitly heuristic; its purpose is to make precise an analogy that we believe is a productive lens, and to state the questions it suggests.

7.1. What a K-Orbit Looks Like

A T -orbit, plotted against iteration count, is the familiar “hailstone” picture: an erratic positive series of surges and crashes (Figure 4). The same orbit transported to K -coordinates (Figure 5) is qualitatively different: it oscillates across zero—because, by Theorem 5, crossing zero is exactly transitioning between the residue classes $[1]_3$ and $[2]_3$ —while its amplitude decays, in fits and starts, toward the sustained terminal oscillation $1, -1, 1, -1, \dots$.

Three quantitative statements attach to this picture.

Proposition 6 (Oscillation frequency). *Along any K -orbit, a step $n \rightarrow K(n)$ preserves the sign if and only if n is negative and even. Under the standard stochastic heuristic [4], in which each iterate is odd or even with probability $\frac{1}{2}$ independently of its sign, the three branch types—positive input, negative odd input, negative even input—form a Markov chain with transition rule*

$$P \rightarrow \{N_{\text{odd}}, N_{\text{even}}\} \text{ (prob. } \frac{1}{2} \text{ each)}, \quad N_{\text{odd}} \rightarrow P, \quad N_{\text{even}} \rightarrow \{N_{\text{odd}}, N_{\text{even}}\} \text{ (prob. } \frac{1}{2} \text{ each)},$$

whose stationary distribution is uniform, $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. The heuristic limiting frequency of sign-changing steps is therefore $2/3$. Empirically, over 2,000 orbits started from uniform random seeds below 10^7 , 66.7% of steps changed sign. Under $\hat{K}$ the frequency is exactly 1: every step changes sign (Theorem 11).

Justification. The first sentence is immediate from (3): the branches for positive input have negative values, the negative-odd branch has positive value, and only the negative-even branch $3n/2$ preserves the sign. Solving $\pi = \pi Q$ for the displayed chain gives $\pi_P = \pi_{N_{\text{odd}}} = \pi_{N_{\text{even}}} = \frac{1}{3}$, and sign-preserving steps occur exactly in state N_{even} . $\square$

Remark 6 (Amplitude drift). *The magnitude multipliers of the four branches of (3) are asymptotically $\frac{3}{2}, \frac{1}{2}, \frac{1}{2}, \frac{3}{2}$ (top to bottom). Under the classical stochastic model of Collatz dynamics [4], in which halving and tripling-halving steps occur independently with probability $\frac{1}{2}$, the expected increment of $\ln |n|$ per step is*

$$-\lambda = \frac{1}{2} \ln \frac{3}{2} + \frac{1}{2} \ln \frac{1}{2} = \frac{1}{2} \ln \frac{3}{4}, \quad \lambda = \frac{1}{2} \ln \frac{4}{3} \approx 0.14384,$$

the same drift as for T (conjugation by J distorts magnitudes only by the bounded factor $|J(n)| \sim n/3$). Empirically, over the same 2,000 random orbits, the mean increment of $\ln |n|$ per K -step was -0.1356 , in reasonable agreement (the discrepancy reflects conditioning on orbits that terminate and the non-equilibrium initial segment). Figure 6 shows $|K^{(t)}(n_0)|$ on a logarithmic scale for several seeds against the envelope $e^{-\lambda t}$.

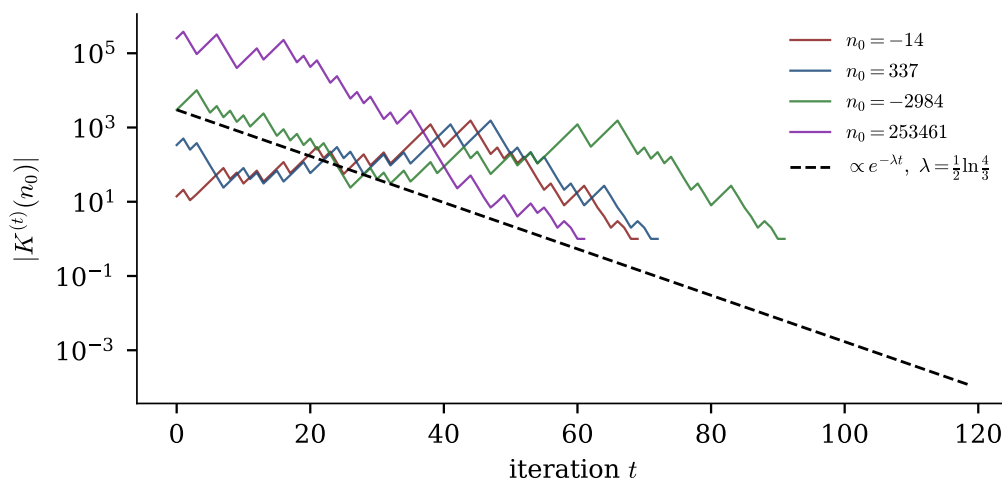


Figure 6. Envelope decay of K -orbits. $|K^{(t)}(n_0)|$ on a log scale for several seeds, against the heuristic envelope $\propto e^{-\lambda t}$ with $\lambda = \frac{1}{2} \ln \frac{4}{3}$ (dashed). Local surges (Remark 5) ride on a reliable exponential decay trend.

7.2. The Seismic Analogy, Stated Carefully

A seismogram of a large earthquake shows a sudden excitation followed by an oscillatory *coda* whose envelope decays essentially exponentially, $A(t) \propto e^{-\omega t/2Q}$, where Q is the coda quality factor [6], until the trace subsides into ambient microseismic oscillation. A K -orbit exhibits the same morphology, with a dictionary:

seismogram	K-orbit
excitation (event)	choice of seed n_0
oscillation about equilibrium	sign changes = $[1]_3 \leftrightarrow [2]_3$ transitions
coda envelope $e^{-\omega t/2Q}$	envelope $e^{-\lambda t}$, $\lambda = \frac{1}{2} \ln \frac{4}{3}$ (heuristic)
irregular waveform	2-adic surges (Remark 5)
ambient noise floor	trivial cycle $1, -1, 1, -1, \dots$
a trace that never subsides	a divergent orbit (conjecturally none)
a self-sustaining resonance	a non-trivial cycle (conjecturally none)

In this language the Collatz conjecture reads: *every excitation decays to the ambient oscillation*—the system is universally damped, admitting neither resonances (non-trivial cycles) nor unbounded responses (divergent orbits). We stress that this is an analogy, not a mechanism: nothing physical underlies it, and by itself it proves nothing. Its value is that it is *quantitatively faithful*—the oscillation is real (Proposition 6), the exponential envelope is real in the almost-everywhere sense made rigorous by Tao’s result that almost all orbits attain almost bounded values [5]—and that it suggests importing the analytical apparatus attached to damped oscillatory signals. The hailstone metaphor, by contrast, suggests no apparatus at all: hailstones fall once.

7.3. Questions This Framing Makes Natural

1. *Spectral statistics.* The sign sequence $\text{sgn}(K^{(t)}(n_0)) \in \{\pm 1\}$ is a deterministic binary signal with empirical switch rate $\approx 2/3$. Do its autocorrelations, or the power spectrum of the normalized signal $K^{(t)}(n_0)/|K^{(t)}(n_0)|^\theta$, distinguish Collatz dynamics from the i.i.d. branch model of [4]? Any provable spectral gap statement would constrain cycle structure.
2. *Decay-rate concentration.* Tao [5] gives almost-everywhere decay along T -orbits. In K -coordinates decay is envelope decay of an oscillation; can one prove concentration of the empirical decay rate $\frac{1}{t} \ln |K^{(t)}(n_0)/n_0|$ around $-\lambda$ on a set of seeds of full density, sharpening the qualitative a.e. statement into a “coda Q ” with error bars?
3. *Resonance exclusion.* In signal terms a non-trivial cycle is a periodic orbit whose geometric mean branch multiplier is exactly 1. The cycle-parity constraint of Corollary 1 and the in-degree arithmetic of Proposition 3 are constraints on how such a resonance could thread the two rays of Figure 2; systematic study of which even-length sign patterns admit integer solutions may be tractable precisely because the sign pattern is now part of the data.
4. *Random dynamical systems.* $\widehat{K}$ alternates deterministically between the rays (Theorem 11), so its second iterate $\widehat{K}^{(2)}$ is a self-map of $\mathbb{Z}^+$ whose fluctuations are governed by the pair (ν_2, ν_3) bookkeeping of Lemma 4. Modeling $\widehat{K}^{(2)}$ as a random dynamical system with explicit multiplier distribution (Proposition 5 supplies the parent-side law) is, we suggest, the natural home for the damping heuristic.

8. Conclusion

We have constructed an explicit conjugacy carrying the conjecture-relevant part of the Collatz graph onto the full set of nonzero integers, and shown that the resulting map K —and its bipartite acceleration $\widehat{K}$ —satisfy Collatz-equivalent conjectures (Theorem 6, Proposition 2) while eliminating the two representational defects of the classical formulations: the inert third of the vertex set, and the unusable negative integers. The reformulation is conservative—every theorem about G_K transports to a theorem about G_{TP} and conversely—but not cosmetic: residue transitions become sign changes, hailstone surges become 2-adic excursions with an explicit growth law, parent multiplicities become the 3-adic valuation $\nu_3(2k - 1)$ with mean $3/2$, and the conjecture itself becomes a damping statement about an oscillatory signal, with a heuristic decay rate $\lambda = \frac{1}{2} \ln \frac{4}{3}$ that measured orbits track closely. We do not expect these coordinates to make the conjecture easy; we do suggest they make some of

its structure—particularly the interplay between the 2-adic and 3-adic bookkeeping and the two-ray geometry of Figure 2—easier to see, and we offer the questions of Section 7 as concrete starting points.

Appendix A Closed Forms and Reference Implementations

The four-branch definition (3) can be written as the single expression

$$K(n) = \frac{1 + \operatorname{sgn}(n)}{2} \left[\varepsilon_n \frac{1 - 3n}{2} - (1 - \varepsilon_n) \frac{n}{2} \right] + \frac{1 - \operatorname{sgn}(n)}{2} \left[\varepsilon_n \frac{1 - n}{2} + (1 - \varepsilon_n) \frac{3n}{2} \right],$$

where $\varepsilon_n = \frac{1 - (-1)^n}{2}$ is the parity indicator.

Python

```
def K(n: int) -> int:
    if n > 0:
        return (1 - 3*n)//2 if n % 2 else -n//2
    return (1 - n)//2 if n % 2 else 3*n//2

def J(n: int) -> int: # [1]3 u [2]3 -> Z*
    return (n + 2)//3 if n % 3 == 1 else -((n + 1)//3)

def J_inv(m: int) -> int: # Z* -> [1]3 u [2]3
    return 3*m - 2 if m > 0 else -3*m - 1

def K_orbit(n: int) -> list[int]:
    orbit = [n]
    while orbit[-1] != 1:
        orbit.append(K(orbit[-1]))
    return orbit
```

Mathematica

```
K[n_Integer?Positive] := If[OddQ[n], (1 - 3 n)/2, -n/2]
K[n_Integer?Negative] := If[OddQ[n], (1 - n)/2, 3 n/2]
KOrbit[n_] := Most@NestWhileList[K, n, # != 1 &]
```

Swift

```
func k(_ n: Int) -> Int {
    switch (n > 0, abs(n % 2) == 1) {
    case (true, true): (1 - 3*n)/2
    case (true, false): -n/2
    case (false, true): (1 - n)/2
    case (false, false): 3*n/2
    }
}

func kOrbit(_ o: [Int]) -> [Int] {
    o.last == 1 ? o : kOrbit(o + [k(o.last!)])
}

/// Convert a K-orbit to the equivalent T-orbit.
func kToT(_ o: [Int]) -> [Int] {
    o.map { $0 > 0 ? 3*$0 - 2 : -3*$0 - 1 }
}

print(kOrbit([20]))
// [20, -10, -15, 8, -4, -6, -9, 5, -7, 4, -2, -3, 2, -1, 1]
print(kToT(kOrbit([20])))
// [58, 29, 44, 22, 11, 17, 26, 13, 20, 10, 5, 8, 4, 2, 1]
```

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